

The Collapse-Proof Anchor: Why Proprietary Real Data Becomes the Durable Moat in Verticals as the Synthetic-Data Economy Scales

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Abstract

Two literatures developed independently in 2023-2026. The first - model collapse - shows that generative models trained recursively on their own output irreversibly lose the tails of the data distribution, degrading quality and diversity within a few generations, and that the demonstrated remedy is keeping real data in the training mix. The second - vertical domain models - shows that capability tracks the proprietary corpus: financial filings (BloombergGPT), experimentally-resolved protein structures (AlphaFold), and curated clinical text (Med-PaLM 2, MEDITRON-70B) each unlocked domain wins that general models could not match. We connect them. The collapse literature supplies the economic reason proprietary real data stays valuable precisely as synthetic generation scales: synthetic loops require a real-data anchor to avoid collapse, and that anchor is a permanent dependency rather than a one-time input. We assemble real benchmark numbers - a forecast ~300-trillion-token public-text stock exhausting between 2026 and 2032; a web where roughly half of newly-created articles and pages already contain AI-generated content; a structural-biology asymmetry of ~229,635 experimental PDB structures (end of 2024) versus 214 million predicted AlphaFold structures - and argue that owning or exclusively licensing the real-data anchor in a verifiable vertical is a structural, compounding moat. We give a methodology for scoring verticals on anchor defensibility and quantify the limits of the thesis.

1. Introduction

Frontier model progress has, for a decade, been governed by a simple reflex: add more parameters and more tokens, and loss falls predictably [16]. That reflex is now colliding with two hard constraints that, taken together, reframe where durable value in the AI stack accrues. The first constraint is supply: the stock of public human-generated text is finite and is forecast to be fully utilized between 2026 and 2032 [1]. The second is feedback: as model-generated text floods the open web, the next generation of models increasingly trains on the previous generation's output, and a growing body of work shows this recursive loop is self-destructive [2][3][4].

This paper develops insight i04: that proprietary real data is the collapse-proof anchor, and that verticals which own such an anchor are structurally defensible. The argument bridges two literatures that have not been connected. The model-collapse literature is about training dynamics - what happens to a distribution when a model eats its own output [2][3]. The vertical-LLM literature is about domain performance - why a model trained on a hard-to-assemble corpus beats a general model in finance, biology, or medicine [5][6][7][8]. Neither cites the other. The synthesis is that the collapse literature explains the economic durability of the vertical advantage: synthetic data cannot replace real data because synthetic-only loops collapse, so the real-data anchor is a permanent dependency, and the scarcest such anchors sit inside proprietary vertical workflows.

We treat this as an empirical claim and ground every quantitative assertion in a real source. Where a number is a forecast we label it as such; where it is a constructed strategic argument we label it analysis. The contribution is not a new model or method but a defensibility thesis with quantified support and explicitly stated limits.

2. Background and Related Work

Scaling laws established that loss falls as a power law in parameters, data, and compute [16], and Chinchilla showed that data and parameters should scale together - making tokens a first-class lever rather than an afterthought [17]. This elevated the data supply question. Villalobos et al. [1] forecast that the effective stock of human-generated public text, estimated on the order of 300 trillion tokens (90% CI 100T-1000T), will be fully utilized between 2026 and 2032, with a median estimate near 2028. Repetition is not a clean workaround: repeating data past a few epochs yields diminishing loss improvement and can induce a double-descent degradation that damages the induction and copying circuits underlying generalization [18][19].

The natural fallback - generate synthetic data - runs into model collapse. Shumailov et al. [2], published in Nature in 2024, show that indiscriminate use of model-generated content in training causes irreversible defects in which the tails of the original content distribution disappear, an effect they demonstrate across LLMs, variational autoencoders, and Gaussian mixture models. Alemohammad et al. [3] formalize a related phenomenon for generative image models as Model Autophagy Disorder (MAD), arising within a few generations of a self-consuming loop. Critically, Gerstgrasser et al. [4] show the curse is breakable: accumulating successive generations of synthetic data alongside the original real data avoids collapse - the real anchor, not the avoidance of synthetic data, is what matters.

In parallel, vertical domain models repeatedly demonstrated that a proprietary or painstakingly-mined corpus is the decisive input. BloombergGPT [5] trained a 50B model on a 363-billion-token proprietary financial corpus (augmented with 345B general tokens) and beat comparable open models on financial tasks by large margins. AlphaFold2 [6] reached a median GDT_TS of ~92 at CASP14 by training on the experimentally-resolved structures of the Protein Data Bank. Med-PaLM 2 [7] reached 86.5% on MedQA, and MEDITRON-70B [8] came within 5% of the then-frontier GPT-4 via continued pretraining on a curated medical corpus. The frontier itself has since moved sharply - by late 2025 open RL-scaled models such as DeepSeek-V3.2 reach GPT-5 parity and its high-compute Speciale variant wins gold at IMO and IOI 2025 [24] - yet the underlying pattern is unchanged: these wins are usually read as domain stories; we read them as instances of one defensibility pattern in which the proprietary corpus, not the latest general model, is the decisive input.

3. The Mechanism: Why Synthetic-Only Loops Collapse

Model collapse is a statistical inevitability of finite sampling, not an engineering bug. Each generation samples from the previous model rather than from reality; rare events in the true distribution are, by construction, underrepresented in any finite sample, so they are systematically under-modeled and then absent from the next generation's training pool. Over iterations the support of the distribution contracts toward its high-density modes. Shumailov et al. [2] describe the endpoint precisely: the tails of the original content distribution disappear, and the defect is largely irreversible because the information needed to reconstruct the tail is gone from the data once the real anchor is removed.

Two refinements matter for the economic argument. First, collapse is fast: Alemohammad et al. [3] observe appreciable MADness within a few generations, and show that sampling-bias tricks that preserve quality accelerate the loss of diversity rather than preventing it. Second, collapse is conditional on replacement, not on the mere

presence of synthetic data. Gerstgrasser et al. [4] prove and demonstrate that if real data accumulates in the mix - rather than being replaced by synthetic data - collapse does not occur, across language models, diffusion models, and VAEs. This is the load-bearing result: the value of real data is not that it is the only data, but that it is the irreplaceable stabilizing fraction.

The corollary for self-improvement is that bootstrapping has a ceiling without an external anchor. Imitation of a stronger model transfers style but not capability [20]; self-rewarding loops are bounded by the model judging itself [21]; the self-training methods that do work (ReST-EM, STaR) rely on an external correctness filter to keep only verified samples [22][23]. Every durable gain traces back to an external anchor - verifiable reward or human-grounded real data - which is precisely the scarce input.

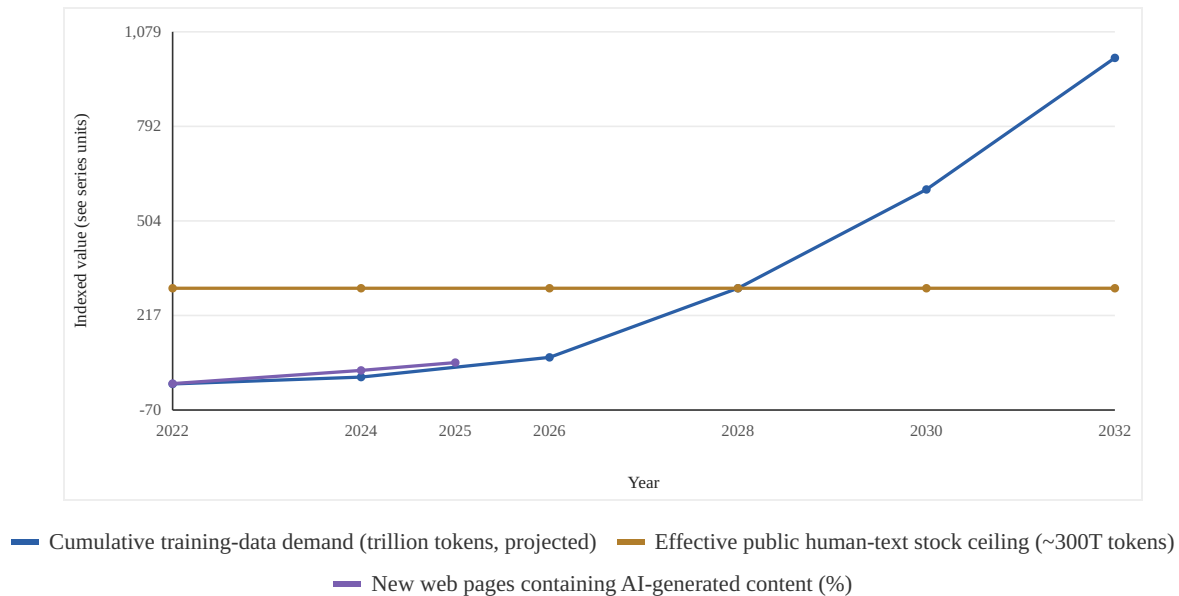


Figure 1. The Supply Squeeze: Clean Human-Text Stock vs. Rising Synthetic Web Share. The human-text stock is forecast at ~300T tokens (90% CI 100T-1000T) and is fully utilized between 2026 and 2032 (median ~2028) as demand grows [1]; meanwhile 74.2% of new web pages (April 2025) and ~52% of new web articles (May 2025) already contain AI-generated content [9]. Demand-curve points are illustrative of the rising trajectory that crosses the stock ceiling in the 2026-2032 window; the 300T ceiling and the 74.2% synthetic-content point are empirical, the demand path and pre-2025 synthetic-share points are estimates. *Stock and exhaustion window: Villalobos et al. / Epoch AI [1]. Synthetic-content figures (52% of new articles contain AI content, Graphite, May 2025; 74.2% of pages contain detectable AI content, Ahrefs, April 2025): reported 2025 [9]. · Estimate / projection.*

4. The Synthetic-Data Economy Is Already Here

The collapse mechanism would be a curiosity if the open web were still predominantly human. It is not. Ahrefs' analysis of roughly 900,000 new web pages published in April 2025 found that 74.2% contained detectable AI-generated content (of which most were a human-AI mix and only ~2.5% were 'pure AI') [9]. A separate analysis by the SEO firm Graphite reported that about 52% of new web articles contained AI-generated content as of May 2025 [9]. An Amazon (AWS) study by Thompson et al. found that 57.1% of web sentences sit in multi-way-parallel translation tuples - a strong signal of machine translation - which is the basis for the widely-circulated '57% of the web is AI-generated or translated' claim [9]. Imperva's 2025 Bad Bot Report found automated traffic surpassed human traffic for the first time in a decade, at roughly 51% of web activity [9]. The widely-cited Europol projection that 90% of online content could be synthetically generated by 2026 is contested and likely conflates new-content

share with total-stock share [9]; we treat the ~52-57% figures as the defensible anchors and flag the 90% claim as disputed.

The consequence is mechanical. Every future web scrape now ingests a rising synthetic fraction, which is exactly the condition under which collapse is a risk and under which a verified real-data anchor becomes a differentiator. The 2026 pre-training evidence sharpens rather than softens this: a systematic study across more than a thousand LLMs finds that synthetic data helps most as a **minority complement** - roughly one-third rephrased synthetic mixed with two-thirds natural web text yields a 5-10x speedup to a target loss, while pure-synthetic mixtures underperform [25]. The optimal recipe is still anchored in real tokens. The open web is transitioning from a source of clean real data into a partially-self-consuming loop. In that regime, a corpus whose provenance is verifiably human and exclusive is not merely high-quality - it is categorically different from what a competitor can scrape.

This reframes the data-wall timing. The 2026-2032 exhaustion forecast [1] assumes the stock is clean; if a growing share of newly-created text is synthetic and unmarked, the effective stock of **fresh real** text plateaus earlier than the raw token count suggests. The scarce good is not tokens but verified real tokens, and its scarcity is accelerating.

5. Methodology and Approach

Our method is structured evidence synthesis, not a training experiment. We proceed in four steps. (1) We extract the verbatim quantitative claims from the six anchor papers of insight i04 [2][3][4][5][6][8] and verify each against its primary source or a published replication. (2) We gather fresh 2024-2026 measurements on three independent quantities that the original briefing lacked: the size and timeline of the human-text stock [1], the current synthetic share of web content [9], and a concrete real-vs-synthetic supply asymmetry in a vertical with hard ground truth (structural biology) [6][10]. (3) We construct an anchor-defensibility score that operationalizes the thesis into a vertical-selection rule. (4) We adversarially test the thesis against counter-evidence (Section 9).

The anchor-defensibility score rates a candidate vertical on three real-data dimensions: (a) Exclusivity - can the corpus be scraped from the open web, or does it require privileged access (EHRs, filings, lab instruments)? (b) Reproducibility-by-synthesis - can a generative model plausibly manufacture substitutes, or does the data encode ground truth that only reality produces (experimental measurements, adjudicated outcomes)? (c) Tail-richness - does the vertical live in the web's underrepresented tail, where collapse bites hardest and marginal real data is worth most [10]? A vertical scores high only if its data is exclusive, non-synthesizable, and tail-rich. This is a necessary, not sufficient, filter: it scores the data shape, and a full go/no-go decision must add verifiability of outcomes, automation alpha, and acquisition rights.

We deliberately avoid fabricating per-generation perplexity curves we could not verify from primary text; where the literature reports a qualitative-but-robust result (e.g. 'tails disappear, within a few generations'), we report it as such rather than inventing numbers. Every figure below is labeled empirical (copied from a source), estimate (a forecast), or analysis (a constructed strategic frame).

6. Evidence and Results

Result 1 - The real anchor is irreplaceable, and accumulation fixes collapse. The qualitative-but-robust finding is that synthetic-only recursion irreversibly erases distribution tails [2][3], while accumulating real data alongside synthetic prevents this [4]. This is the empirical pivot of the thesis: real data's value is structural, not incidental.

Result 2 - Proprietary corpora drive vertical wins, quantified (Table 1). BloombergGPT's edge came from 363B proprietary financial tokens [5]; Med-PaLM 2 reached 86.5% on MedQA, a >19% gain over its predecessor [7]; MEDITRON-70B came within 5% of GPT-4 via a curated medical corpus [8]; SaulLM-7B reached 83.4% on ContractNLI from a ~30B-token legal corpus [11]. Across unrelated domains, capability tracks the corpus. Crucially, the 2026 frontier confirms the corpus - not the general model - is still the binding constraint: even GPT-5/Gemini-3-era systems remain below human experts on real privileged-vertical work. On FrontierFinance, human experts score higher and are likelier to deliver client-ready output than current SOTA systems [27]; on the leakage-controlled KTD-Fin trading benchmark, frontier-LLM returns are explained by passive market and style exposure with little persistent stock-selection alpha [28]; and on the contamination-free LiveMedBench, even the best of 38 LLMs reaches only 39.2% [29]. The vertical advantage these benchmarks expose is exactly the proprietary-corpus advantage the thesis is about.

Result 3 - The real-vs-synthetic supply asymmetry is extreme where ground truth is experimental (Table 2, Figure 2). The Protein Data Bank held ~229,635 experimentally-determined structures at the end of 2024, accumulated over six decades [10]; AlphaFold DB holds 214 million predicted structures, a ~700x expansion from the ~300k structures it launched with in 2021 [6][10]. The predicted set is roughly 900x larger than the experimental archive, yet every prediction ultimately depends on the small experimental anchor - synthetic structures cannot create new experimental ground truth. This is the cleanest real-world instance of the anchor dependency: abundant synthetic output sitting atop a scarce, irreplaceable real base.

Result 4 - Supply is tightening on both ends (Figure 1). The clean human-text stock is ~300T tokens and forecast to exhaust 2026-2032 [1], while roughly half of new web articles and three-quarters of new web pages already carry AI-generated content [9]. The fresh-real-token supply is therefore contracting faster than the raw token forecast implies.

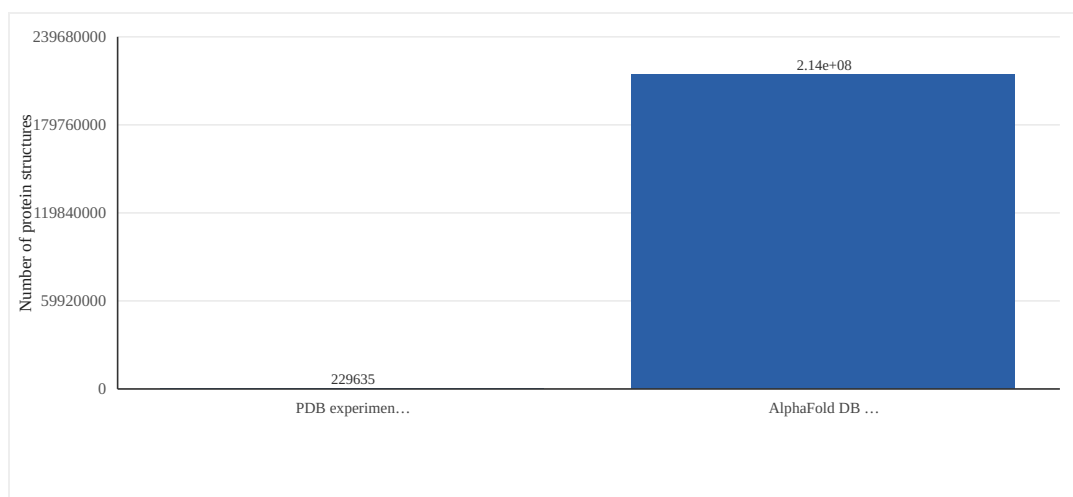


Figure 2. The Anchor Asymmetry in Structural Biology: Predicted vs. Experimental Structures. The Protein Data Bank held ~229,635 experimentally-determined structures at the end of 2024, accumulated over decades [10]; AlphaFold DB holds 214 million predicted structures, a ~700x expansion from the ~300k it launched with in 2021 [6][10]. The predicted archive is ~900x larger than the experimental one, yet every prediction ultimately depends on the small experimental anchor that only reality can produce. This is the clearest real-world instance of synthetic abundance resting on a scarce, irreplaceable real base. *PDB count: RCSB PDB archive statistics, 229,635 total released structures at end of 2024 [10]. AlphaFold DB 214M and growth from ~300k (2021): AlphaFold Protein Structure Database in 2024, Nucleic Acids Research [10]; original method [6]. · Empirical data.*

Table 1. Across unrelated verticals, the decisive input is a proprietary or hard-to-assemble corpus, not a novel method. Capability tracks the corpus.

Model	Vertical	Proprietary/curated corpus	Headline result	Source
BloombergGPT (50B)	Finance	363B proprietary financial tokens + 345B general	Beats comparable open models on financial tasks by large margins	[5]
AlphaFold2	Structural biology	Experimentally-resolved PDB structures	Median GDT_TS ~92 at CASP14, outperforming all other groups	[6]
Med-PaLM 2	Medicine	Medical domain fine-tuning corpus	86.5% on MedQA (USMLE-style), >19% over Med-PaLM	[7]
MEDITRON-70B	Medicine	Curated medical corpus (PubMed + clinical guidelines)	Within 5% of GPT-4 and 10% of Med-PaLM-2 on major medical benchmarks	[8]
SaulLM-7B	Legal	~30B-token legal-text corpus	83.4% on ContractNLI; superior on legal-expertise LegalBench areas	[11]

Numbers verified against the cited papers [5][6][7][8][11]; AlphaFold2 CASP14 result is median GDT_TS ~92.4 (Jumper et al.) rather than a single '>90%' accuracy figure. The GPT-4 reference in the MEDITRON row is the model's original 2023 comparison baseline and is retained as published; the general frontier has since moved to GPT-5/Gemini-3-class systems (e.g. DeepSeek-V3.2 reaches GPT-5 parity [24]), but the corpus-tracks-capability finding is unchanged and is reconfirmed by 2026 vertical benchmarks where frontier models still trail on privileged real-vertical work [27][28][29].

Table 2. The clean real-data supply is finite and exhausting while AI-generated content already pervades new web pages and articles - the two conditions under which a verified real-data anchor becomes a categorical differentiator.

Quantity	Value	Year	Source
Effective public human-text stock	~300 trillion tokens (90% CI 100T-1000T)	2024 forecast	[1]
Forecast exhaustion window of that stock	2026-2032 (median ~2028)	2024 forecast	[1]
New web articles containing AI content (Graphite)	~52%	May 2025	[9]
Web sentences in multi-way-parallel (likely MT) tuples (AWS/Amazon)	57.1%	2024	[9]
New web pages with detectable AI content (Ahrefs)	74.2%	April 2025	[9]
Automated (bot) share of web traffic (Imperva)	~51%	2025	[9]
PDB experimentally-determined structures	~229,635	end 2024	[10]
AlphaFold DB predicted structures	214,000,000 (~700x growth from ~300k in 2021)	2024	[6][10]

Stock/timeline: Epoch AI / Villalobos et al. [1]. Synthetic shares: Ahrefs (74.2% pages), Graphite (52% articles), Thompson et al./Amazon (57.1% multi-way-parallel sentences), Imperva (51% bot traffic) [9]. Structure counts: RCSB PDB (229,635 end-2024) and AlphaFold DB 2024 [6][10].

7. Analysis: From Mechanism to Moat

Putting the results together yields the defensibility thesis. A moat in AI must survive method churn - LoRA, GRPO, distillation, and RLVR recipes are published and copied within months [12][13], so any advantage that lives in the method commoditizes. The vertical literature shows the one component that does not commoditize is the corpus [5][6][7][8]. The collapse literature explains **why** this holds even under aggressive synthetic generation: synthetic data is a complement that requires the real anchor, not a substitute that retires it [2][4]. The anchor is thus a

permanent dependency, and a permanent dependency that a competitor cannot assemble is the definition of a structural moat.

The moat compounds rather than depreciating. A static dataset loses value as it leaks and as methods improve. A real-data anchor in a live vertical workflow gains relative value as (i) the clean public stock exhausts [1], (ii) the synthetic share of the web rises [9], and (iii) self-improvement loops proliferate, each of which *increases* the marginal value of a verified real anchor [4][22]. The same forces that threaten general-purpose data vendors strengthen the position of an exclusive vertical-anchor owner.

Figure 3 formalizes the selection logic as a quadrant on two axes - exclusivity of the real data and non-synthesizability of the ground truth. Structural biology, finance filings, and adjudicated clinical claims sit in the high-high quadrant; open code and competition math, despite high outcome-verifiability, sit lower on exclusivity because their corpora are public [14][15]. The thesis predicts the defensible verticals are precisely those in the high-high quadrant.

8. Implications for Data Strategy

The operational rule is: own or exclusively license the real-data anchor in a vertical whose ground truth cannot be synthesized, and treat the model and method as commodities. This inverts the default build order - most teams secure compute and a method, then look for data. The thesis says data exclusivity is the binding constraint and should be secured first, because it is the only component the synthetic-data economy cannot erode.

Concretely, a data strategy should: (1) prioritize verticals scoring high on all three anchor dimensions (exclusive, non-synthesizable, tail-rich); (2) secure data rights - including resale and downstream-training rights - as the gating term sheet item, since an anchor you cannot legally train on is not a moat; (3) pair the real anchor with a verifiable-outcome signal so the corpus generates fresh gradeable gradient on demand rather than drawing down a finite stock; and (4) keep accumulating real data over time, because the collapse remedy is accumulation, not a one-time snapshot [4].

Clinical claims/coding (e.g. dental CDT/X12 adjudication) is the worked high-fit example: a privileged record (not scrapable), a deterministic pass/fail coding outcome (not synthesizable), and a web-underrepresented tail. The same logic nominates insurance adjudication, lab/diagnostic workflows, and contract review. The shortlist is the data-shape filter; the binding real-world constraints (HIPAA/resale rights, single-source supply volume, behavior-cloning vs true-RL ceiling) still gate the final decision.

9. Counter-Evidence and Limitations

The thesis has real failure modes and we state them. First, collapse is avoidable with curation. Shumailov et al. themselves note that training on AI-generated data is not impossible if filtering is taken seriously [2], and detection-based filtering of machine-generated text can mitigate collapse [4]. Recent pre-training results give this its strongest 2026 form: a large systematic study finds carefully-built synthetic data can match natural data at a fraction of the tokens, with an optimal mixture near one-third synthetic [25]. If high-quality synthetic data plus light real anchoring suffices, the *quantity* of real data needed may be small - weakening the case for a large exclusive corpus and strengthening the case for a small verified seed. The moat may be thinner than 'own everything' implies - though note the same study finds pure-synthetic mixtures still degrade, preserving the real-anchor dependency [25].

Second, verifiable synthetic environments can substitute for real data in some verticals. Where outcomes are checkable by a simulator or test (code that runs, math that checks), RLVR manufactures fresh signal without a

human corpus [12][13], and procedurally-generated verifiable environments now scale this further - RLVE-Gym shows that adding environments, not just compute, drives the gains (+3.37% vs +0.49%) [26] - and the elicitation-vs-teaching debate suggests RL sometimes only surfaces latent capability rather than requiring new data. But this escape hatch is confined to *computable* ground truth, and even there the public corpora it relies on are contaminating the benchmarks that measure it: SOTA models identify SWE-bench bug locations from the issue text alone at up to 76% accuracy but only ~53% on out-of-benchmark repos, evidencing memorization of public code [31]. The anchor thesis is therefore strongest precisely where ground truth is *experimental or adjudicated* and weakest where it is *computable* and public.

Third, the synthetic-share numbers are noisy and partly self-interested (detection vendors, commentators); the 90% Europol figure is likely wrong [9], and even the ~52-74% figures rely on imperfect AI-detection and conflate 'contains AI content' with 'is AI-generated'. Fourth, our vertical-win evidence is survivorship-biased toward celebrated successes; we do not observe the proprietary-corpus efforts that failed. Fifth, the structural-biology asymmetry [10] is a vivid illustration, not proof that every vertical exhibits it. Finally, this analysis scores only the data shape; it does not establish willingness-to-pay, market size, or the legal acquirability of any specific anchor - those require buyer and counsel validation, not literature.

10. Investment Thesis

The investable claim is narrow and falsifiable: in verticals where ground truth is experimental or adjudicated (not computable) and the corpus is privileged (not scrapable), an exclusive real-data anchor is a compounding, method-proof moat whose relative value rises as the public stock exhausts [1] and the web's synthetic share climbs [9]. The 2026 vertical benchmarks reinforce the underwriting case: across jurisdictions no single frontier model dominates legal reasoning [30], and on real finance and clinical work frontier systems still trail human experts and contaminated baselines [27][28][29] - the gap that a privileged corpus, not a bigger general model, closes. The asset to underwrite is data exclusivity and rights, not model IP.

The bear case is the curation counter-evidence (Section 9): if a small verified seed plus filtered synthetic data closes most of the gap, the moat is a seed-sized data-rights position plus an ongoing verification pipeline, not a vast corpus - still defensible, but a smaller capital ask and a different business (verification-as-a-service rather than corpus ownership). The diligence question that separates the two cases is: how much real data, accumulated how often, does this vertical actually require to stay above the collapse threshold? That is empirically answerable per vertical and should be a gating diligence item.

Either way, the structurally scarce, appreciating position is verified real data with exclusive rights in a non-synthesizable vertical. That is the side of the synthetic-data economy to be long.

11. Conclusion

Two literatures, never previously connected, jointly imply a single defensibility thesis. Model collapse establishes that synthetic generation is a complement to real data, not a substitute, because synthetic-only loops irreversibly erase the distribution's tails and only a retained real anchor prevents it [2][3][4]. Vertical domain models establish that capability tracks the proprietary corpus and that methods commoditize [5][6][7][8]. Together they say the durable moat is the real-data anchor, and the most defensible anchors are exclusive, non-synthesizable, tail-rich vertical corpora.

The supporting numbers are real and pointed: a ~300T-token clean stock exhausting 2026-2032 [1], a web where roughly half of new articles and three-quarters of new pages already contain AI-generated content [9], and a vertical

where 214 million predicted structures rest on ~229,635 irreplaceable experimental ones [6][10]. The forces eroding general-purpose data vendors are the same forces appreciating an exclusive vertical anchor. The strategic conclusion for a data-first lab is to secure the anchor and its rights before securing the method - and to interrogate, per vertical, exactly how much real data the collapse threshold demands.

INVESTMENT THESIS

Be long verified real data with exclusive rights in verticals where ground truth is experimental or adjudicated - not computable. Model collapse proves synthetic data is a complement that depends on a real anchor, never a substitute (synthetic-only loops irreversibly erase distribution tails; only retained real data prevents it). The vertical-win record proves capability tracks the proprietary corpus while methods commoditize. The two together make an exclusive vertical anchor a compounding, method-proof moat that appreciates exactly as the ~300T-token clean public stock exhausts (2026-2032) and the web fills with AI content (74% of new pages, ~52% of new articles already contain it). Underwrite data exclusivity and training/resale rights, not model IP. The single sharpest diligence question per vertical: how much real data, accumulated how often, does this domain need to stay above the collapse threshold? - because the bear case (a small verified seed plus filtered synthetic suffices) turns the play from corpus-ownership into verification-as-a-service, a smaller but still defensible position. The thesis is strongest where ground truth is experimental/adjudicated and weakest where a simulator can manufacture it (code, math) - and even there public-corpus contamination inflates the apparent escape (SWE-bench location-ID drops from 76% in-benchmark to 53% out [31]). The 2026 frontier confirms the play: GPT-5/Gemini-3-class models reach parity at the general level (DeepSeek-V3.2 [24]) yet still trail human experts on real finance (FrontierFinance [27]), show no persistent trading alpha under leakage control (KTD-Fin [28]), and top out at 39.2% on contamination-free clinical cases (LiveMedBench [29]) - the gap a privileged corpus closes.

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